

Guidelines for Labeling and Annotation of Planetary Geological Maps

1General Principles

This study targets a YOLO-based detection task for map layout elements. The model aims to stably and accurately locate functional information regions rather than perform pixel-level semantic understanding.

Therefore, the label definition follows the core principle of being dominated by layout structure and supplemented by semantic information. Specifically:

- (1) Category division is prioritized based on spatial position, visual structure, and layout stability.
- (2) Consistency and learnability of the detected targets in visual appearance are pursued.

a. map_frame

Definition

- (1) Closely follow the outer boundary of the actual drawn area of the map.
- (2) Do not actively include latitude/longitude numerical annotations.

Permissible Circumstances

For circular or irregular map frames, if the inclusion of a small number of latitude/longitude numerical annotations is unavoidable, they may be incorporated into the `map_frame`.

The principle is to adhere as closely as possible to the main body of the map rather than striving for complete exclusion of text.

b. Legend

Case1:Tabular Legend

Treat the entire table (including the table header) as a single `legend` box. Do not split it into individual rows or columns.

Case2:Long-Text Legend

Must conform to one of the following structures:

(1) Graphic + Text description

(2) Title + Graphic + Text description

Case3:Dispersed Legend

If the legend items form a visually coherent whole in space (commonly seen on the right side of a map), they may be enclosed as a whole with a rectangular box.

If individual legend items are clearly separated from the main group, they may be annotated as independent `legend` boxes.

c. Title

Definition

The definition of `title` prioritizes the integrity of overall expression over individual lines or blocks of text.

Case 1: Standard Title

Located at the top or bottom of the entire map;

Unique and clearly expresses the map's theme;

Directly annotated as `title`.

Case 2: Irregularly Dispersed Titles (e.g., AIC / LTO / Lunar Chart series)

Title-related information is distributed across multiple locations (e.g., upper-left, lower-left, lower-right corners);

All text blocks that constitute the map's naming and numbering system are uniformly annotated as a single `title`;

Even if such blocks contain scale bar or year information, they are not further subdivided.

Note:

In this map style, an independent `scale block` already exists. Therefore, the presence of scale information within a `title` does not affect downstream parsing.

d. diagram

Definition

Includes auxiliary graphics such as cross-sections, statistical charts, axes, unit labels, and any charts other than the map itself.

Does not include textual explanations of the diagram (e.g., "Figure 1: This diagram shows..."). Such explanatory text shall be classified under `metadata_block`.

e. north_arrow

Annotate as an independent block;

Closely follow the edges of the symbol;

Do not include surrounding descriptive text.

f. scale_block

Definition

A separately enclosed area containing scale information, which may appear in any of the following forms:(1) Numerical scale (e.g., 1:50,000);(2) Bar scale;(3) Combination of numerical scale + bar scale;

Design Note

A standalone "1:50,000" text, if scattered in isolation, can easily be treated as noise by the model.

Therefore, only scale regions with stable structure and visual compactness are annotated.

In downstream processing, the specific value will be parsed through a combination of OCR and computer vision.

Special Note

During label formulation, priority is given to annotation based on the page layout structure.

h. metadata_block (元数据文本)

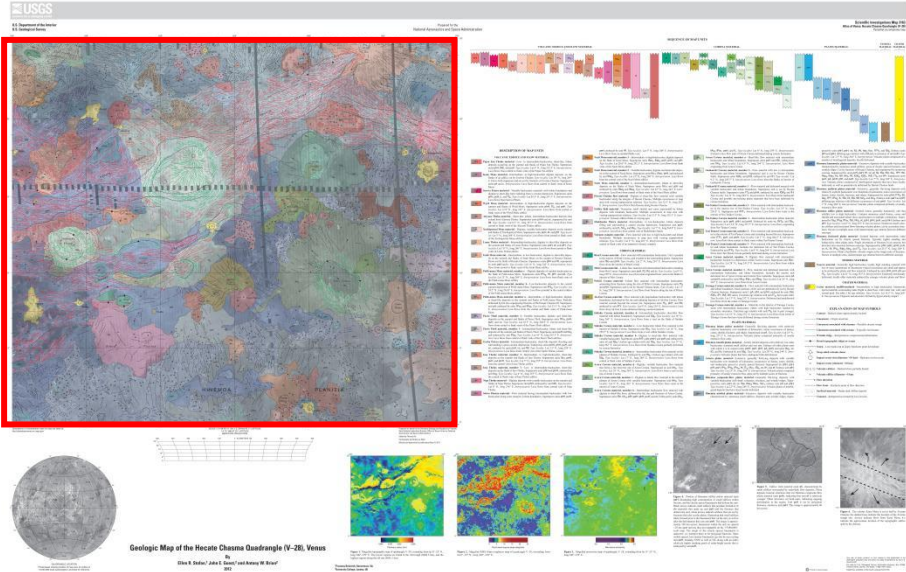
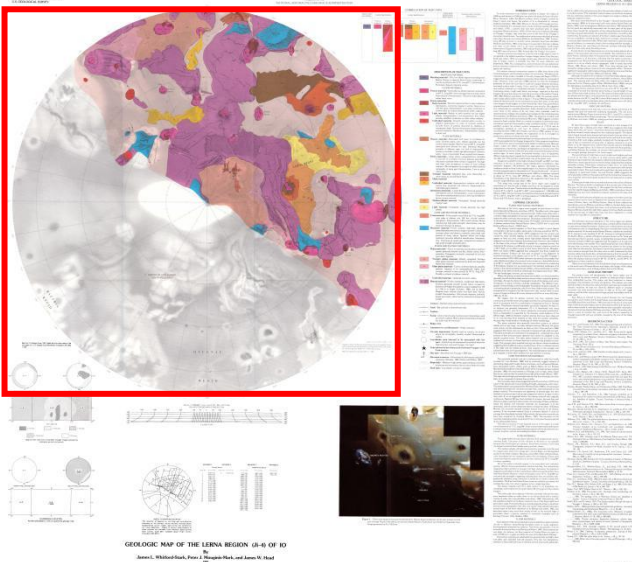
Definition

All textual information not covered by the six categories above, including but not limited to:Data source\Projection description\Explanatory paragraphs

2.Principles for Spatial Intersection and Inclusion

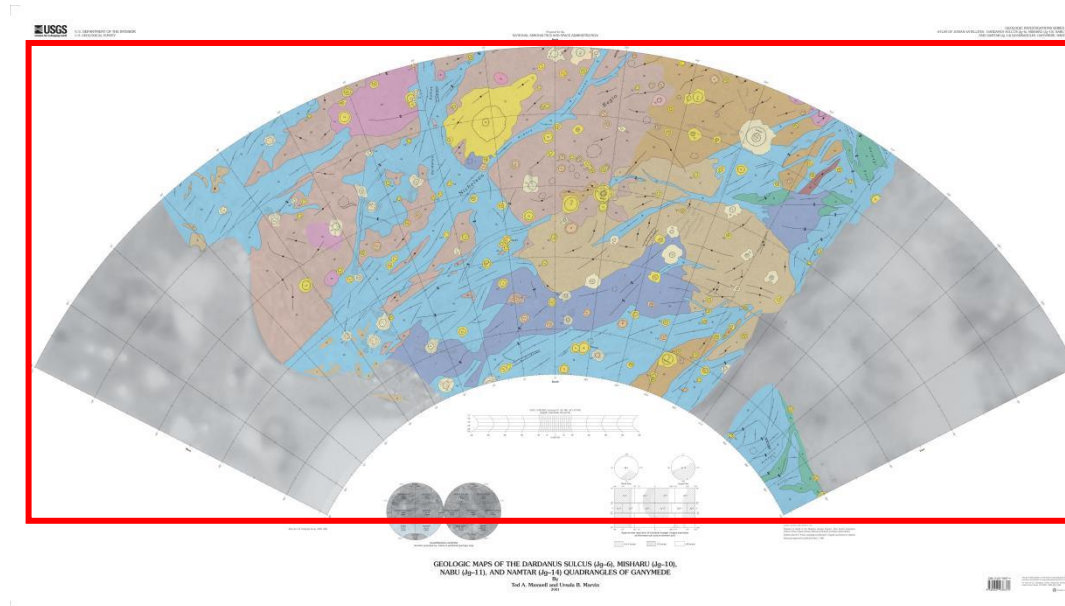
Partial spatial intersection between annotation boxes of different categories is permitted; Complete overlap (high IoU 叠置) between two annotation boxes of different categories is not permitted;

When functional elements (legend, north_arrow, scale_block) lie inside the map_frame, they shall still be annotated independently as separate boxes; the map_frame shall retain its full boundary and shall not be cropped due to internal elements.

Category	Example	Annotation Guidelines	Notes
1.1 Rectangle Box		Definition: The boundary line of a rectangular map frame, used to delineate the map area.	Adhere to the main body of the map frame, excluding text or numerical annotations adjacent to the map frame.
1.map_frame 1.2 Circular box		Definition: The boundary line of a circular map frame, used to delineate the map area.	(1) Adhere as closely as possible to the main body of the map. (2) Inevitable inclusion of edge numbers is permitted. (3) Do not deliberately shrink

the box to exclude
such numbers.

1.3
irregular
box



Definition: An irregular map frame resulting from different projection methods or clipping, used to delineate the map area.

(1) Adhere as closely as possible to the main body of the map.

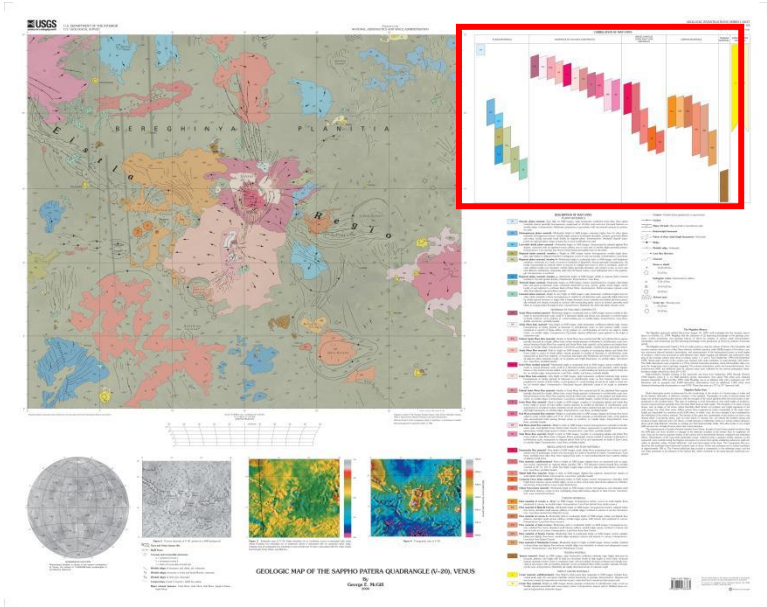
(2) Inevitable inclusion of edge numbers is permitted.

(3) Do not deliberately shrink the box to exclude such numbers.

2.diagram

2.1

color chart

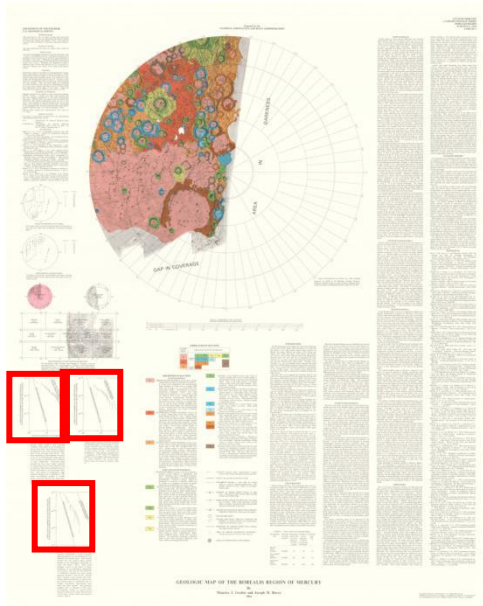


Definition: A chart that uses colors to represent the relationships between different geological units.

Include all elements of the complete chart, including the title and category names.

2.2

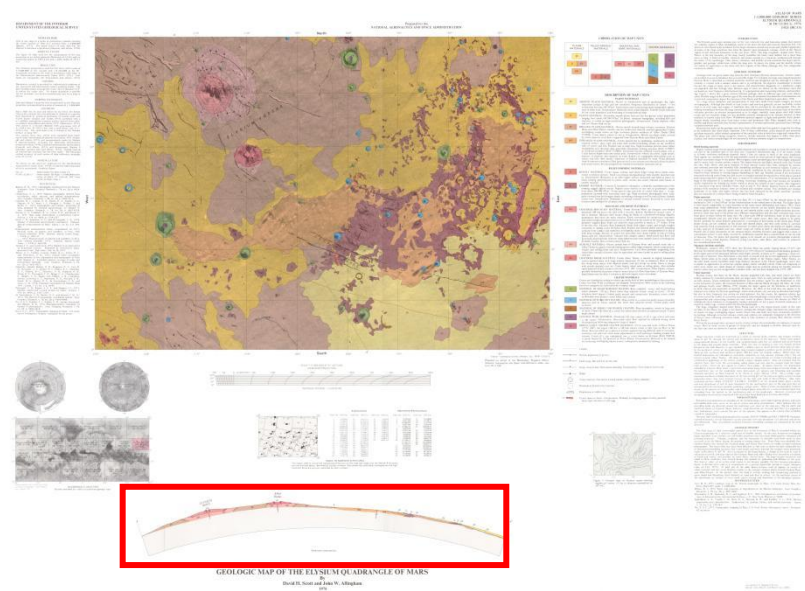
line chart



Definition: A line chart is an auxiliary diagram used to display the trend of one or more variables along a continuous coordinate axis. Its core visual characteristic consists of one or more trend lines formed by a series of data points connected by line segments.

Include x-axis and y-axis labels.

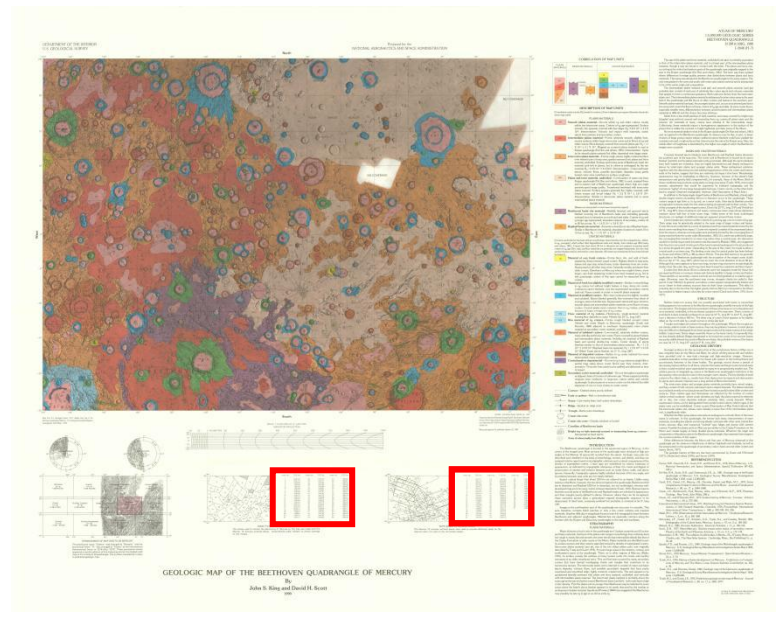
2.3
cross-
section



Definition: A diagram showing
geological structures and
stratigraphic sequences
underground or along a
specific line.

Include x-axis and
y-axis labels.

2.4
number list



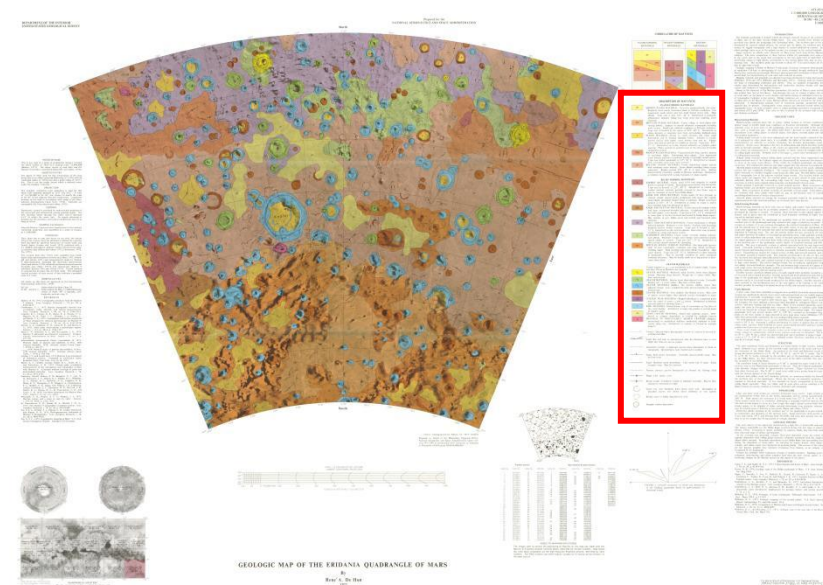
Definition: A list of Arabic
numerals arranged in a
concentrated manner in a vertical
or horizontal sequence, used for
sequential identification or
indexing..

Include the
chart/diagram title
and category
names.

3.legend

3.1

textual
legend

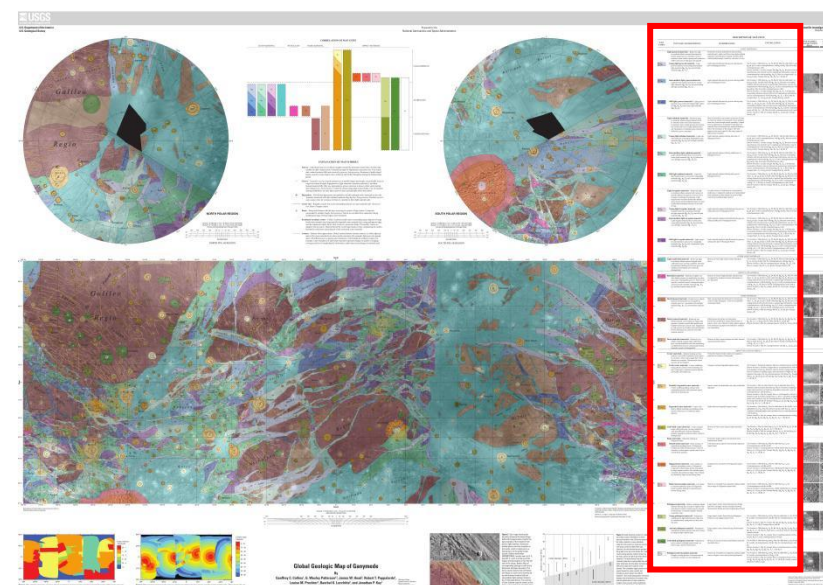


Definition: A legend
description presented in the
form of a long strip of text.

The legend area
must contain clear
cartographic
symbols (e.g.,
colors, patterns, line
types, icons) along
with their
corresponding
semantic
descriptions.

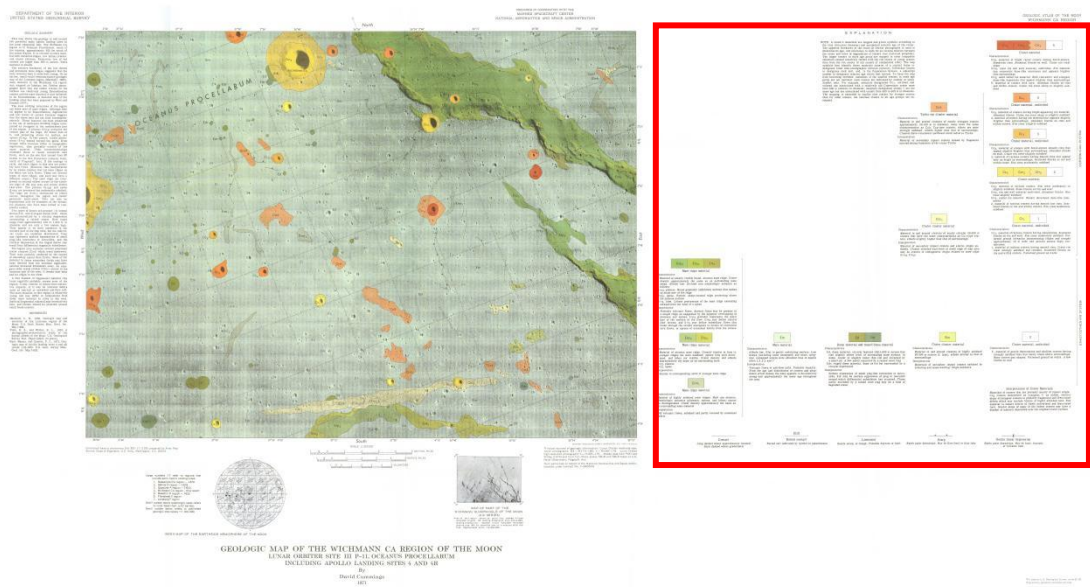
3.2

tabular
legend



Definition: A legend descriptio
n presented in the form of a
tabular text.

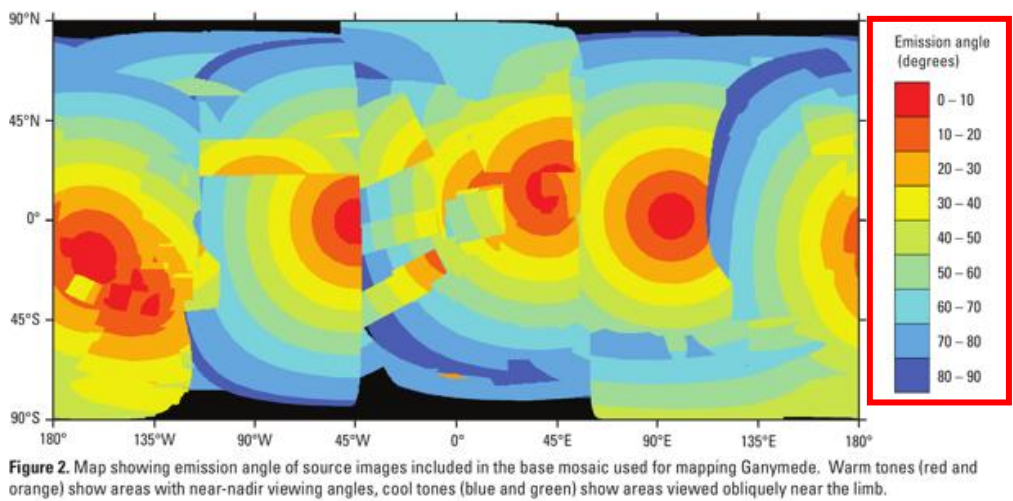
3.3
irregularly
dispersed
legend



Definition: A legend arranged
in an irregular manner.

- (1) Draw a bounding
box around the
entire region
containing the
legend.
- (2) If this would
cause substantial
overlap with
another region,
partially segment
and annotate
separately.

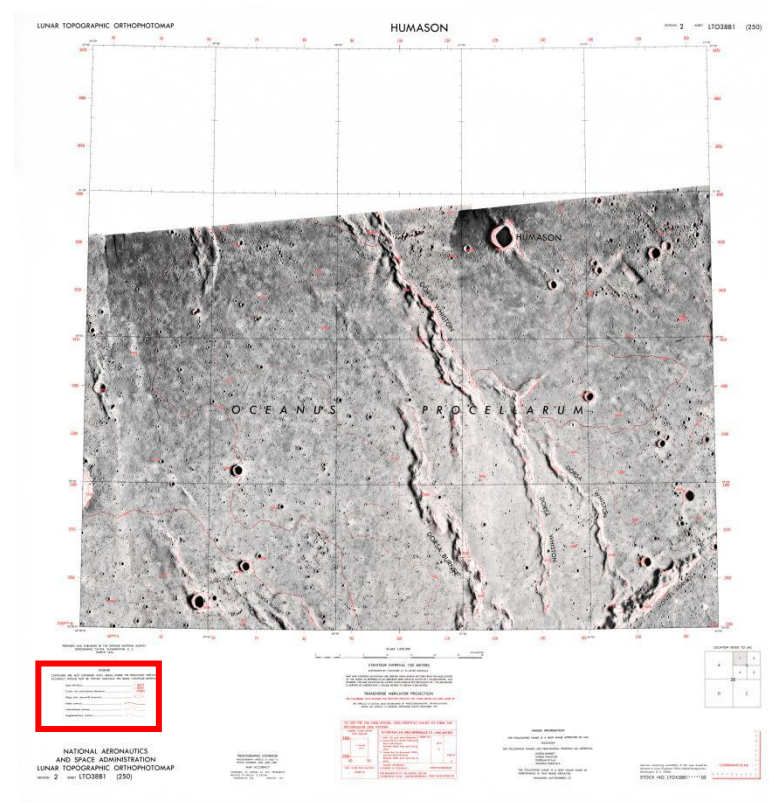
3.4
Hypsometr-
ic legend



Definition: A legend block
consisting of color patches,
where different colors
represent different meanings.

- (1) Include
the legend title.
- (2) Numerical
annotations should
also be included.

3.5
contour
legend
with
numerical
annotation



Definition: A legend that
represents elevation and
contour lines.

Mark the entire
region as a whole.

3.6			Definition: A color strip with a	(1) Include the
continuous			continuous gradient and	legend title.
gradient			numerical tick marks (or scale).	(2) Numerical
legend				annotations shall
				also be included.
The image is a topographic map of the Moon's South Pole region, titled "Topographic Map of the Moon's South Pole (85°S to Pole)". It uses a polar stereographic projection with a scale of 1:1,343,325. The map shows various craters and features, including Cabeus, Haworth, Shoomaker, Malinlin, Faustini, de Gerlache, Shackleton, Sverdrup, Kuhn, Wiedert J, Wiedert P, Slator, Nobite, Metapert, and Scott M. A color-coded elevation scale is provided in the bottom left, ranging from -5480 meters (low, purple) to 7010 meters (high, yellow). The map is framed by latitude and longitude lines (0°E, 90°E, 90°W, 180°E). The map is credited to the USRA LPI Regional Planetary Image Facility, Lunar and Planetary Institute, Houston, Texas, USA. Data sources are listed as Lunar Reconnaissance Orbiter Lunar Orbiter Laser Altimeter (LOLA) 20-m elevation product [NASA Goddard Space Flight Center] over derived hillshade with solar azimuth 45°W and solar elevation 45°. LPI Contribution No. 2171, May 2019.				
4.metadata_ block	Text that does not belong to the mandatory textual content of the other six categories shall be classified as metadata_block.			

5.title
5.1
standard
title



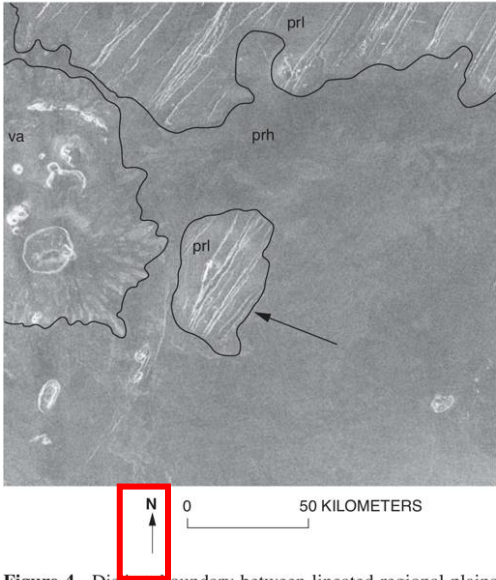
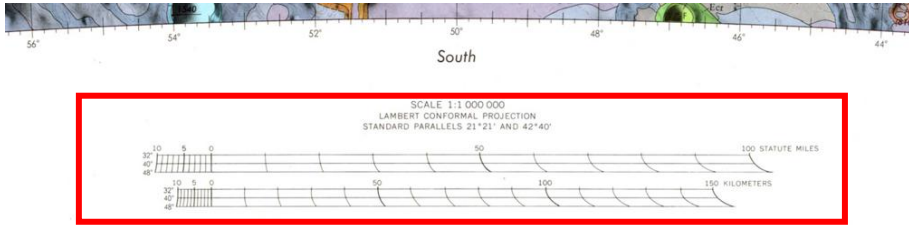
Definition: A main title
indicating the theme of the
current map, which in this case
also includes the author and
date.

5.2
scattered
title



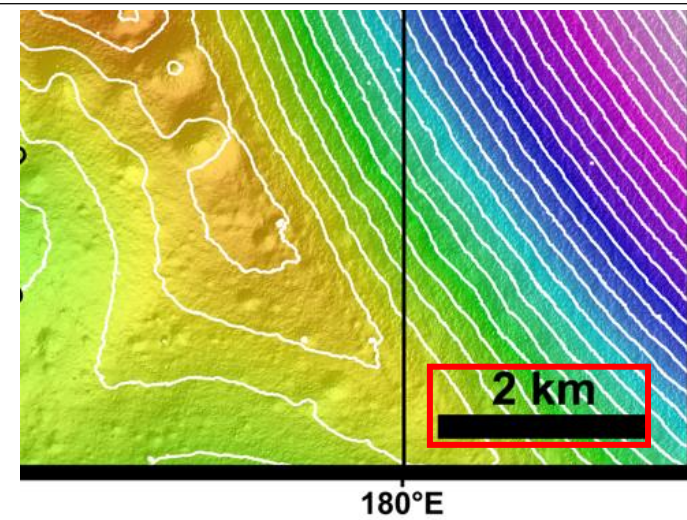
Definition: A title that requires
selecting multiple scattered
boxes to convey the subject.

This situation
typically occurs in
lunar atlases or
maps of the
following styles:
(1) Apollo
Intermediate Charts
(AIC),
(2) Lunar Chart
(LAC) Series,
(3) Lunar Map (LM)
Series,
(4) Lunar Shaded

				Relief (LSR) Series, (5) Ranger Lunar Charts (RLCs).
6.north arrow		 Figure 4. Distinct boundary between lineated regional plains material (unit prl), which contains a strong fabric of grabens and lineaments, and homogeneous regional plains material (unit prh). Arrow points to a kipuka of unit prl within unit prh (also see fig. 5). Unit va is anemone material. Image is centered at 34° S., 16° E.	Definition: north arrow	Must consist of an arrow accompanied by a letter label.
7. Scale block	7.1 text + projection information + scale bar graphic		Definition: A region representing scale information, typically including scale text, projection information, and a scale bar.	When a scale text (e.g., 1:5,000,000) appears alone, it shall not be annotated separately. Such text often also appears within the title area;

in this case, it
should not be
separated into a
distinct scale block
but rather be
incorporated into
the title region.

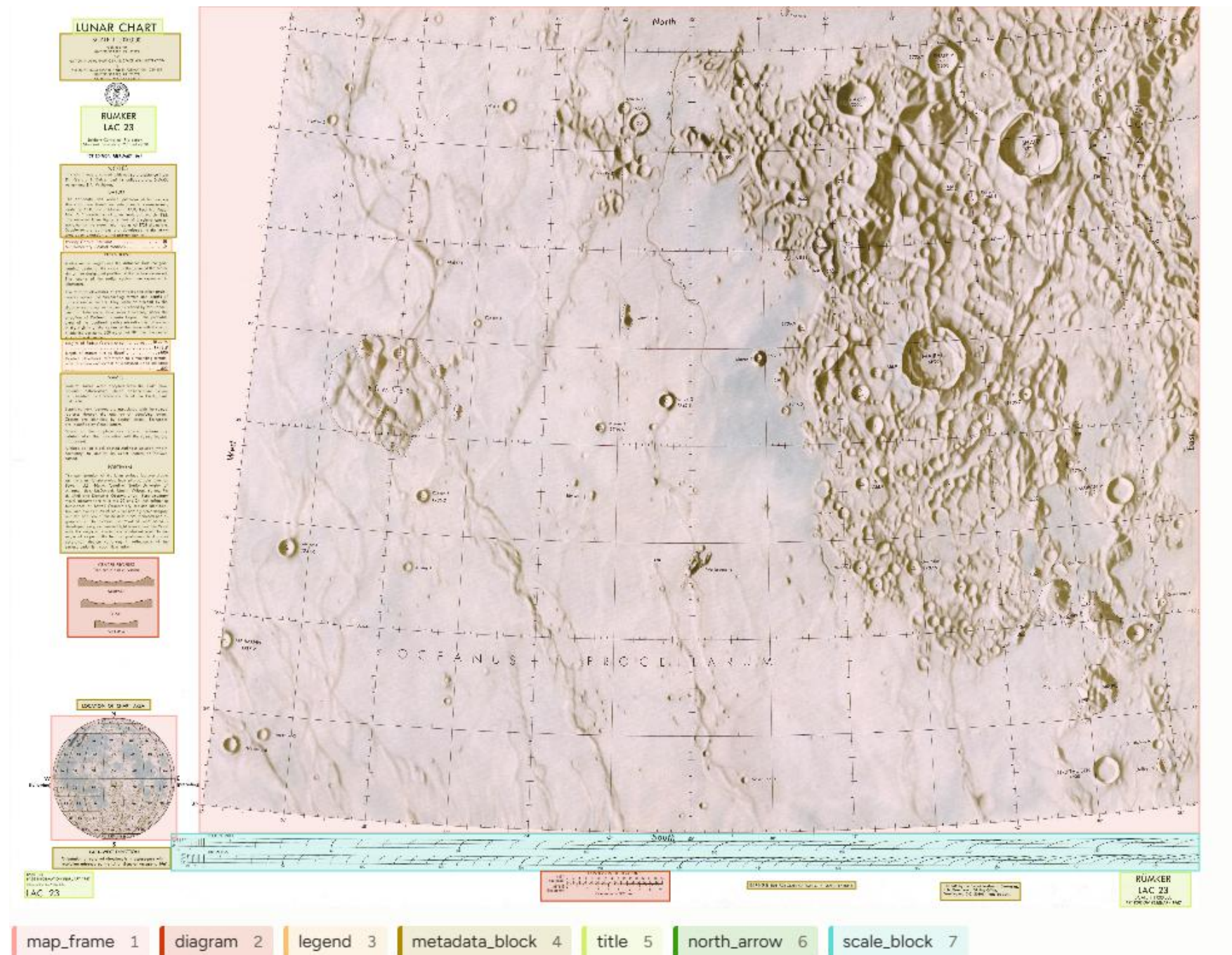
7.2
scale bar
graphic

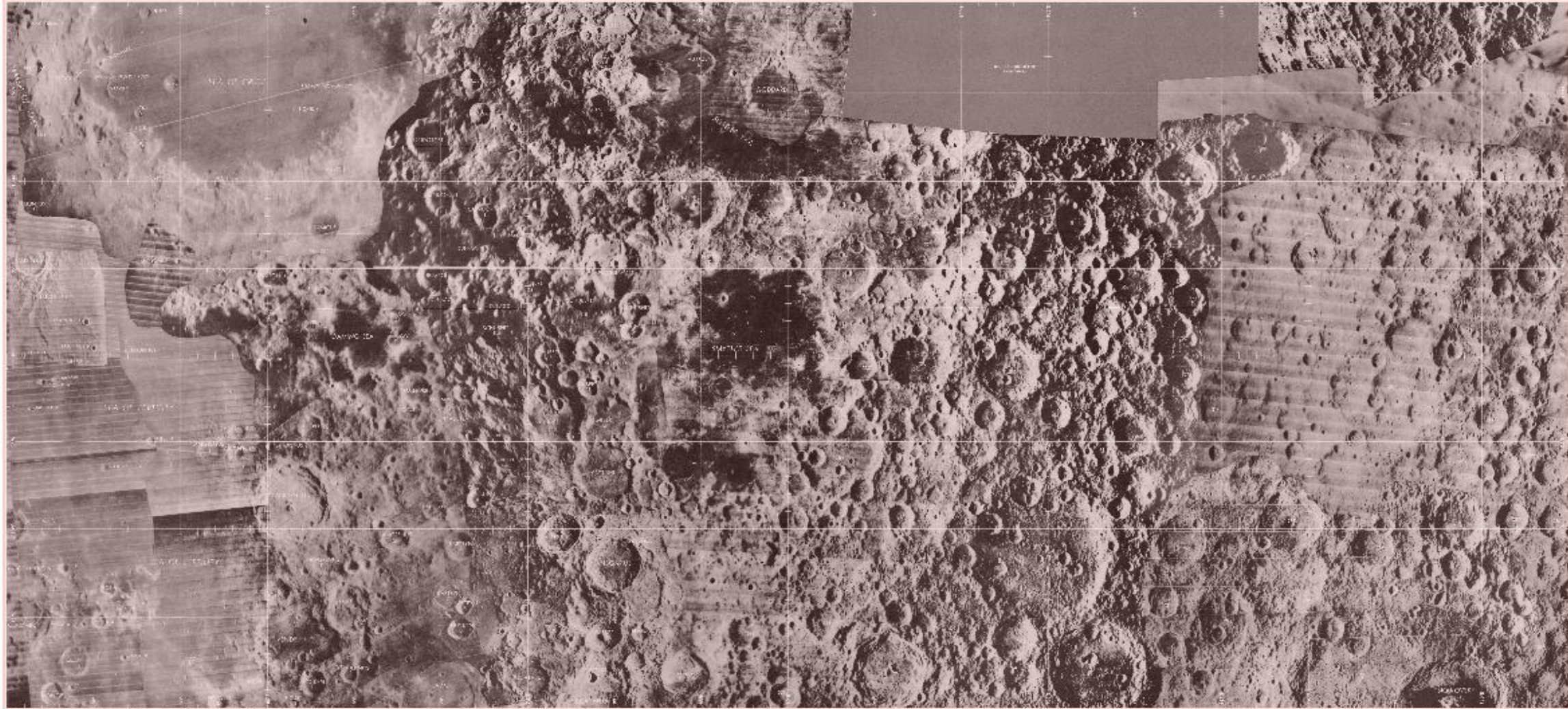


Data Sources:
Lunar Reconnaissance Orbiter Lunar Orbiter Laser Altimeter (LOLA)
10 m elevation product [NASA Goddard Space Flight Center] over
shaded hillshade with solar azimuth 45°W and solar elevation 45°.
High elevation: red and orange; Low elevation: purple and white.

Definition: Numerical unit
annotations on a scale bar that
convey scale information.

Annotation Examples for Different Map Styles:



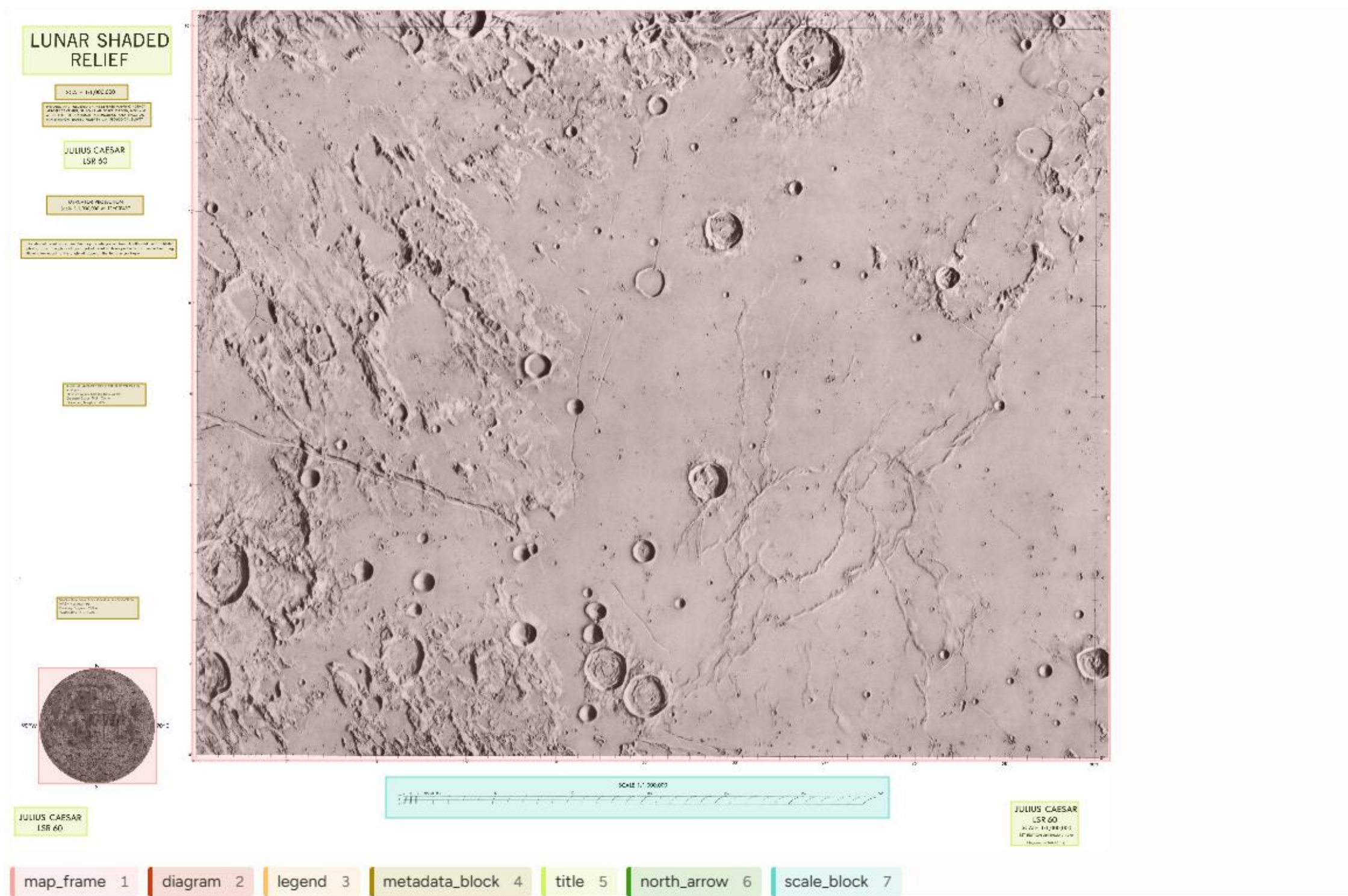


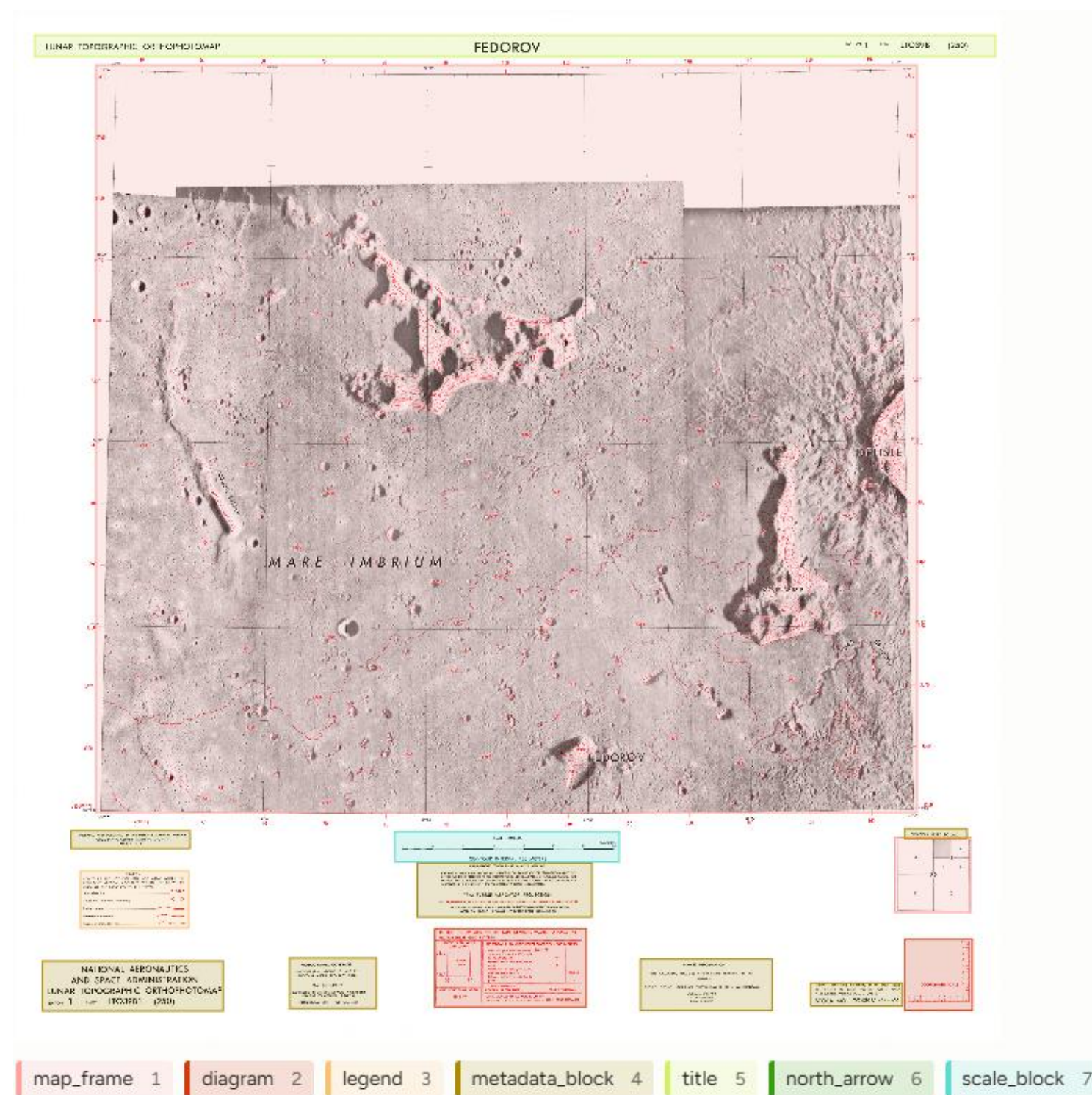
LUNAR COORDINATE ZONE A6546
W 100000
EASTERN 1000

UNITED STATES GEOLOGICAL SURVEY
BULLETIN 1452
WASHINGTON, D. C. 20508



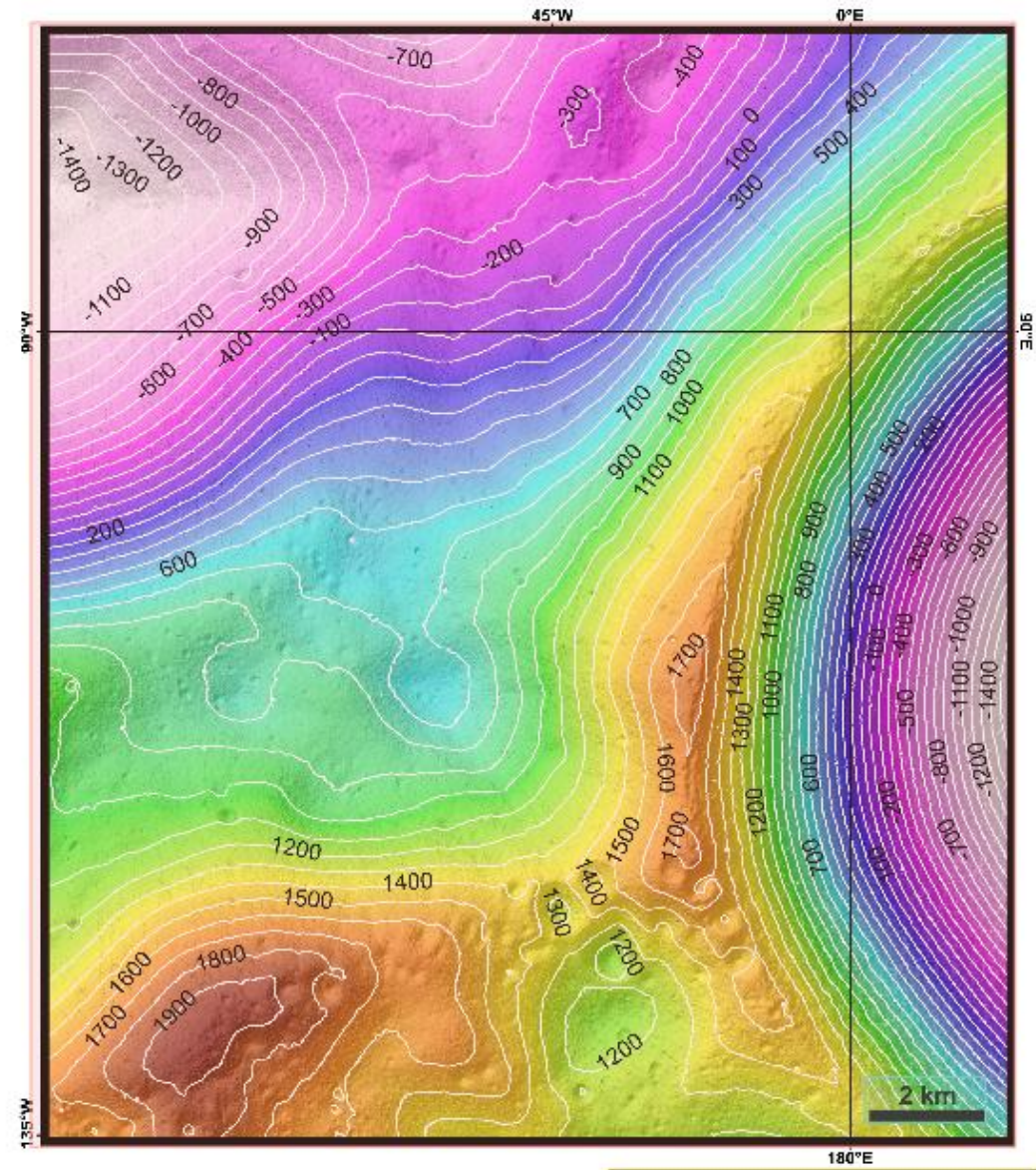
map_frame 1 | diagram 2 | legend 3 | metadata_block 4 | title 5 | north_arrow 6 | scale_block 7





Topographic Contour Map of the Moon's South Pole Ridge

Pseudocylindrical Projection
Contours: 100 m



Center for Lunar Science and Exploration
Lunar and Planetary Institute
Houston, Texas, USA

Data Sources:
Lunar Reconnaissance Orbiter Lunar Orbiter Laser Altimeter (LOLA)
5-m elevation product (NASA/Goddard Space Flight Center, open-
derivative hillshade with color azimuth 45°W and color elevation 45°).
High elevation: red and orange; Low elevation: purple and white.

map_frame 1

diagram 2

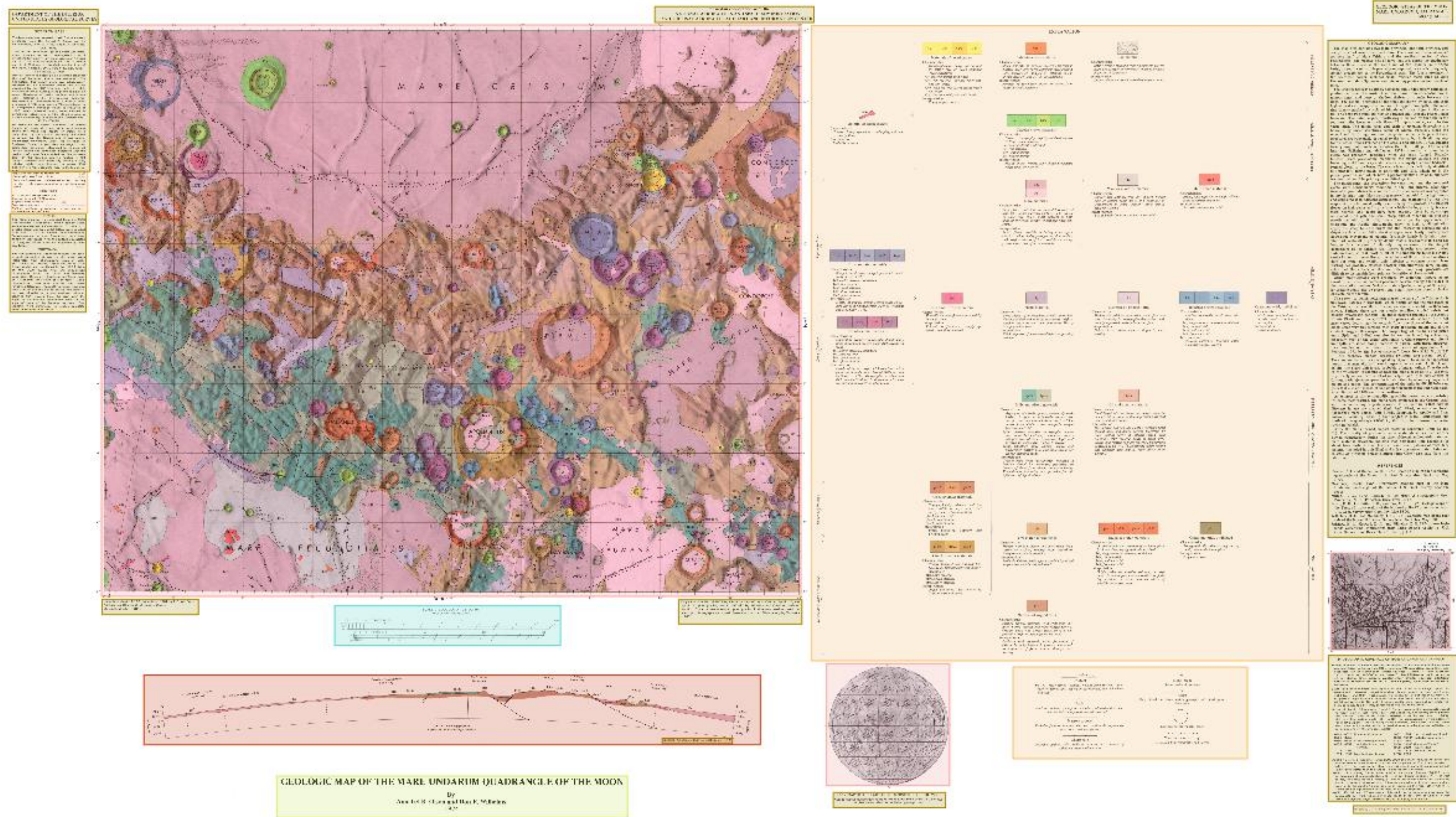
legend 3

metadata_block 4

title 5

north_arrow 6

scale_block 7



map_frame 1 | diagram 2 | legend 3 | metadata_block 4 | title 5 | north_arrow 6 | scale_block 7